

My First

SQL Case Study

For E-Commerce Retail Business

Global Electronics

Global Electronics

Is a Retailer which runs a worldwide company to sell different electronics products for its customer.

Objectives

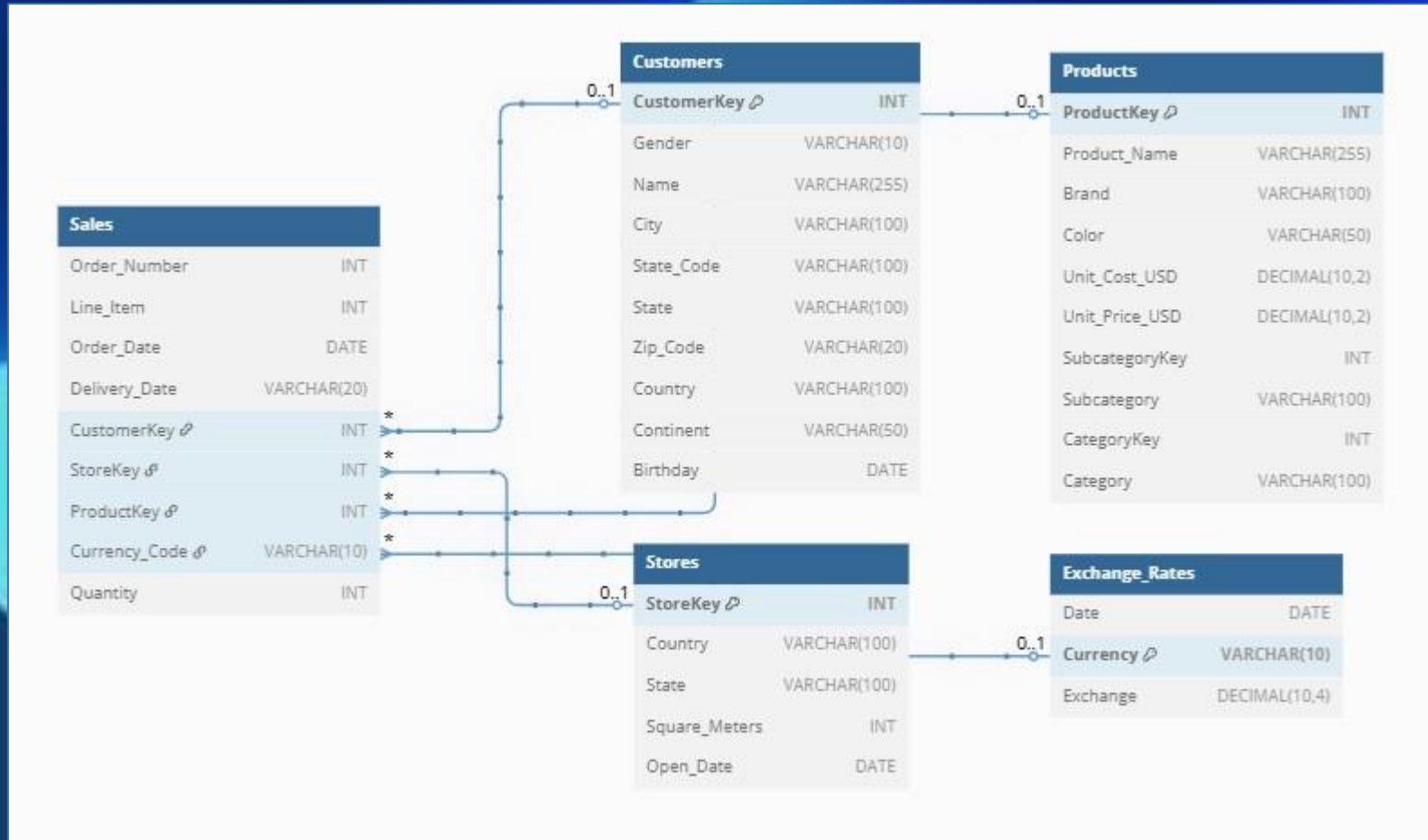
- Sales and Revenue analysis.
- Customer behavior analysis
- Product and Delivery performance
- RFM Segmentations and Cohort analysis

Business Goal

- Find out most contributing products in revenue
- seasonal patterns or trends for order volume or revenue
- Average delivery time & order value

What I did here,

At first, I get the dataset from Maven Analytics. Then create a database in MySQL.
Here is the schema model;



Dataset Overview

Sales →

	Order_Number	Line_Item	Order_Date	Delivery_Date	CustomerKey	StoreKey	ProductKey	Quantity	Currency_Code
▶	366000	1	2016-01-01		265598	10	1304	1	CAD
	366001	1	2016-01-01	2016-01-13	1269051	67	1048	2	USD
	366001	2	2016-01-01	2016-01-13	1269051	67	2007	1	USD
	366002	1	2016-01-01	2016-01-12	266019	67	1106	7	CAD

Customer →

	CustomerKey	Gender	Name	City	State_Code	State	Zip_Code	Country	Continent	Birthday
▶	301	Female	Lilly Harding	WANDEARAH EAST	SA	South Australia	5523	Australia	Australia	1939-07-03
	325	Female	Madison Hull	MOUNT BUDD	WA	Western Australia	6522	Australia	Australia	1979-09-27
	554	Female	Claire Ferres	WINJALLOK	VIC	Victoria	3380	Australia	Australia	1947-05-26

Products →

	ProductKey	Product_Name	Brand	Color	Unit_Cost_USD	Unit_Price_USD	SubcategoryKey	Subcat	Cateç	Category
▶	1	Contoso 512MB MP3 Player E51 Silver	Contoso	Silver	6.62	12.99	101	MP4...	1	Audio
	2	Contoso 512MB MP3 Player E51 Blue	Contoso	Blue	6.62	12.99	101	MP4...	1	Audio
	3	Contoso 1G MP3 Player E100 White	Contoso	White	7.40	14.52	101	MP4...	1	Audio

Store →

	StoreKey	Country	State	Square_Meters	Open_Date
▶	1	Australia	Australian Capital Territory	595	2008-01-01
	2	Australia	Northern Territory	665	2008-01-12
	3	Australia	South Australia	2000	2012-01-07
	4	Australia	Tasmania	2000	2010-01-01

Exchange Rates →

	Date	Currency	Exchange
▶	2015-01-01	AUD	1.2214
	2015-01-02	AUD	1.2323
	2015-01-03	AUD	1.2323

Key Metrics

Total Revenue

```
-- 1. TOTAL REVENUE
SELECT SUM(QUANTITY * UNIT_PRICE_USD) AS TOTAL_REVENUE
FROM SALES
JOIN PRODUCTS ON SALES.PRODUCTKEY = PRODUCTS.PRODUCTKEY;
```

	TOTAL_REVENUE
▶	55755479.59

Total Profit

```
-- TOTAL PROFIT
SELECT SUM(QUANTITY *(UNIT_PRICE_USD - UNIT_COST_USD)) AS TOTAL_PROFIT
FROM SALES
JOIN PRODUCTS ON SALES.PRODUCTKEY = PRODUCTS.PRODUCTKEY;
```

	TOTAL_PROFIT
▶	32662688.38

Total Products

```
-- TOTAL PRODUCTS
SELECT COUNT(DISTINCT PRODUCTKEY) AS TOTAL_PRODUCTS
FROM PRODUCTS;
```

	TOTAL_PRODUCTS
▶	2517

Total Customers

```
-- 3. TOTAL CUSTOMERS
SELECT COUNT(DISTINCT CUSTOMERKEY) AS TOTAL_CUSTOMERS
FROM CUSTOMERS;
```

	TOTAL_CUSTOMERS
▶	15266

Total Orders

```
-- TOTAL ORDERS
SELECT COUNT(DISTINCT ORDER_NUMBER) AS TOTAL_ORDERS
FROM SALES;
```

	TOTAL_ORDERS
▶	26326

Total Quantity Sold

```
-- TOTAL QUANTITY SOLD
SELECT SUM(QUANTITY) AS TOTAL_QUANTITY_SOLD
FROM SALES;
```

	TOTAL_QUANTITY_SOLD
▶	197757

Revenue Trends

Aggregates revenue by year

```
-- YEARLY REVENUE
SELECT YEAR(ORDER_Date) AS YEAR, SUM(QUANTITY * UNIT_PRICE_USD) AS YEARLY_REVENUE
FROM SALES
JOIN PRODUCTS ON SALES.PRODUCTKEY = PRODUCTS.PRODUCTKEY
GROUP BY YEAR
ORDER BY YEAR;
```

	YEAR	YEARLY_REVENUE
▶	2016	6946793.56
	2017	7421422.27
	2018	12788960.66
	2019	18264382.48
	2020	9294632.14
	2021	1039288.48

Highest revenue generates in 2019

Aggregates revenue by years & months

```
-- MONTHLY REVENUE
SELECT YEAR(ORDER_DATE) AS YEAR, MONTH(ORDER_Date) AS MONTH,
      SUM(QUANTITY * UNIT_PRICE_USD) AS MONTHLY_REVENUE
FROM SALES
JOIN PRODUCTS ON SALES.PRODUCTKEY = PRODUCTS.PRODUCTKEY
GROUP BY YEAR, MONTH
ORDER BY YEAR, MONTH;
```

YEAR	MONTH	MONTHLY_REVENUE
2016	1	649918.78
2016	2	891098.30
2016	3	338407.36
2016	4	110591.63
2016	5	595986.18
2016	6	518585.28
2016	7	454959.83
2016	8	547512.69
2016	9	567421.12
2016	10	623959.57
2016	11	649949.36
2016	12	998403.46

YEAR	MONTH	MONTHLY_REVENUE
2017	1	670114.80
2017	2	750526.19
2017	3	331248.59
2017	4	58580.07
2017	5	616258.96
2017	6	631731.30
2017	7	511786.72
2017	8	593379.05
2017	9	686375.10
2017	10	628521.43
2017	11	655722.99
2017	12	1287177.07

YEAR	MONTH	MONTHLY_REVENUE
2018	1	917075.35
2018	2	1344342.70
2018	3	434951.15
2018	4	70627.54
2018	5	1073265.56
2018	6	973975.33
2018	7	964149.46
2018	8	1026845.73
2018	9	1181765.14
2018	10	1241730.03
2018	11	1478026.63
2018	12	2082206.04

YEAR	MONTH	MONTHLY_REVENUE
2019	1	1940899.57
2019	2	2102862.57
2019	3	845925.09
2019	4	149892.71
2019	5	1594446.47
2019	6	1404861.40
2019	7	1408714.62
2019	8	1500784.77
2019	9	1547870.73
2019	10	1575168.82
2019	11	1715659.88
2019	12	2477295.85

YEAR	MONTH	MONTHLY_REVENUE
2020	1	2068951.12
2020	2	2227379.57
2020	3	674990.66
2020	4	217642.10
2020	5	878026.63
2020	6	763883.23
2020	7	512805.18
2020	8	416647.08
2020	9	380431.52
2020	10	245647.59
2020	11	256701.02
2020	12	651526.44

YEAR	MONTH	MONTHLY_REVENUE
2021	1	513021.58
2021	2	526266.90

Month Over Month Revenue Change (%)

```
WITH MONTHLY_REVENUE AS (  
    SELECT YEAR(ORDER_DATE) AS YEAR, MONTH(ORDER_DATE) AS MONTH,  
    SUM(QUANTITY * UNIT_PRICE_USD) AS REVENUE  
    FROM SALES  
    JOIN PRODUCTS ON SALES.PRODUCTKEY = PRODUCTS.PRODUCTKEY  
    GROUP BY YEAR,MONTH)  
SELECT YEAR, MONTH,  
    REVENUE,  
    REVENUE - COALESCE(LAG(REVENUE) OVER (ORDER BY YEAR, MONTH),0) AS MOM_CHANGE,  
    ROUND(CASE  
        WHEN LAG(REVENUE) OVER (ORDER BY YEAR, MONTH) IS NULL THEN NULL  
        ELSE (REVENUE - LAG(REVENUE) OVER (ORDER BY YEAR, MONTH)) * 100 / LAG(REVENUE) OVER (ORDER BY YEAR, MONTH)  
        END, 2) AS MOM_PERCENTAGE_CHANGE  
FROM MONTHLY_REVENUE;
```

Last 2 years result 2019,2020 with running year 2021

YEAR	MONTH	REVENUE	MOM_CHANGE	MOM_PERCENTAGE_CHANGE
2020	1	2068951.12	-408344.73	-16.48
2020	2	2227379.57	158428.45	7.66
2020	3	674990.66	-1552388.91	-69.70
2020	4	217642.10	-457348.56	-67.76
2020	5	878026.63	660384.53	303.43
2020	6	763883.23	-114143.40	-13.00
2020	7	512805.18	-251078.05	-32.87
2020	8	416647.08	-96158.10	-18.75
2020	9	380431.52	-36215.56	-8.69
2020	10	245647.59	-134783.93	-35.43
2020	11	256701.02	11053.43	4.50
2020	12	651526.44	394825.42	153.81

YEAR	MONTH	REVENUE	MOM_CHANGE	MOM_PERCENTAGE_CHANGE
2019	1	1940899.57	-141306.47	-6.79
2019	2	2102862.57	161963.00	8.34
2019	3	845925.09	-1256937.48	-59.77
2019	4	149892.71	-696032.38	-82.28
2019	5	1594446.47	1444553.76	963.73
2019	6	1404861.40	-189585.07	-11.89
2019	7	1408714.62	3853.22	0.27
2019	8	1500784.77	92070.15	6.54
2019	9	1547870.73	47085.96	3.14
2019	10	1575168.82	27298.09	1.76
2019	11	1715659.88	140491.06	8.92
2019	12	2477295.85	761635.97	44.39

YEAR	MONTH	REVENUE	MOM_CHANGE	MOM_PERCENTAGE_CHANGE
2021	1	513021.58	-138504.86	-21.26
2021	2	526266.90	13245.32	2.58

Profit Trends

```
-- YEARLY PROFIT
SELECT YEAR(ORDER_Date) AS YEAR, SUM(QUANTITY *(UNIT_PRICE_USD - UNIT_COST_USD)) AS YEARLY_PROFIT
FROM SALES
JOIN PRODUCTS ON SALES.PRODUCTKEY = PRODUCTS.PRODUCTKEY
GROUP BY YEAR
ORDER BY YEAR;
```

YEAR	YEARLY_PROFIT
2016	4107000.47
2017	4337064.06
2018	7464961.11
2019	10697738.90
2020	5447460.15
2021	608463.69

Highest in 2019

```
-- MONTHLY PROFIT
SELECT YEAR(ORDER_Date) AS YEAR, MONTH(ORDER_DATE) AS MONTH, SUM(QUANTITY *(UNIT_PRICE_USD - UNIT_COST_USD)) AS YEARLY_PROFIT
FROM SALES
JOIN PRODUCTS ON SALES.PRODUCTKEY = PRODUCTS.PRODUCTKEY
GROUP BY YEAR, MONTH
ORDER BY YEAR, MONTH;
```

Last 2 years result 2019,2020
with running year 2021

YEAR	MONTH	YEARLY_PROFIT
2019	1	1136859.17
2019	2	1237893.26
2019	3	496096.95
2019	4	88472.45
2019	5	933498.44
2019	6	833108.11
2019	7	821894.29
2019	8	877158.96
2019	9	906551.14
2019	10	915753.53
2019	11	1014904.36
2019	12	1435548.24

YEAR	MONTH	YEARLY_PROFIT
2020	1	1209119.39
2020	2	1311449.23
2020	3	396204.76
2020	4	128029.54
2020	5	514047.95
2020	6	443405.96
2020	7	302322.37
2020	8	245161.15
2020	9	222284.38
2020	10	141098.51
2020	11	149679.25
2020	12	384657.66

YEAR	MONTH	YEARLY_PROFIT
2021	1	300479.71
2021	2	307983.98

Month Over Month Profit Change (%)

```
-- MONTH ON MONTH PROFIT CHANGE
```

```
WITH MONTHLY_PROFIT AS (
```

```
    SELECT YEAR(ORDER_DATE) AS YEAR, MONTH(ORDER_DATE) AS MONTH,  
    (SUM(QUANTITY * (UNIT_PRICE_USD - UNIT_COST_USD))) AS PROFIT  
    FROM SALES
```

```
    JOIN PRODUCTS ON SALES.PRODUCTKEY = PRODUCTS.PRODUCTKEY
```

```
    GROUP BY YEAR,MONTH)
```

```
SELECT YEAR, MONTH,
```

```
    PROFIT,
```

```
    PROFIT - COALESCE(LAG(PROFIT) OVER (ORDER BY YEAR, MONTH),0) AS MOM_CHANGE,
```

```
    ROUND(CASE
```

```
        WHEN LAG(PROFIT) OVER (ORDER BY YEAR, MONTH) IS NULL THEN NULL
```

```
        ELSE (PROFIT - LAG(PROFIT) OVER (ORDER BY YEAR, MONTH)) * 100 / LAG(PROFIT) OVER (ORDER BY YEAR, MONTH)
```

```
        END, 2) AS MOM_PERCENTAGE_CHANGE
```

```
FROM MONTHLY_PROFIT;
```

YEAR	MONTH	PROFIT	MOM_CHANGE	MOM_PERCENTAGE_CHANGE
2019	1	1136859.17	-66701.05	-5.54
2019	2	1237893.26	101034.09	8.89
2019	3	496096.95	-741796.31	-59.92
2019	4	88472.45	-407624.50	-82.17
2019	5	933498.44	845025.99	955.13
2019	6	833108.11	-100390.33	-10.75
2019	7	821894.29	-11213.82	-1.35
2019	8	877158.96	55264.67	6.72
2019	9	906551.14	29392.18	3.35
2019	10	915753.53	9202.39	1.02
2019	11	1014904.36	99150.83	10.83
2019	12	1435548.24	420643.88	41.45

YEAR	MONTH	PROFIT	MOM_CHANGE	MOM_PERCENTAGE_CHANGE
2020	1	1209119.39	-226428.85	-15.77
2020	2	1311449.23	102329.84	8.46
2020	3	396204.76	-915244.47	-69.79
2020	4	128029.54	-268175.22	-67.69
2020	5	514047.95	386018.41	301.51
2020	6	443405.96	-70641.99	-13.74
2020	7	302322.37	-141083.59	-31.82
2020	8	245161.15	-57161.22	-18.91
2020	9	222284.38	-22876.77	-9.33
2020	10	141098.51	-81185.87	-36.52
2020	11	149679.25	8580.74	6.08
2020	12	384657.66	234978.41	156.99

YEAR	MONTH	PROFIT	MOM_CHANGE	MOM_PERCENTAGE_CHANGE
2021	1	300479.71	-84177.95	-21.88
2021	2	307983.98	7504.27	2.50

Last 2 years result 2019,2020 with running year 2021

The Biggest Drops are seen in March and April

Product Analysis

Products Category

CATEGORY
Audio
TV and Video
Computers
Cameras and camcorders
Cell phones
Music, Movies and Audio Books
Games and Toys
Home Appliances

Top 10 Product Sub-Category by Revenue

```
-- Top 10 Product Sub-Categories by Revenue --  
SELECT SUBCATEGORY,  
       SUM(QUANTITY * UNIT_PRICE_USD) AS TOTAL_REVENUE  
FROM PRODUCTS  
JOIN SALES ON PRODUCTS.PRODUCTKEY = SALES.PRODUCTKEY  
GROUP BY SUBCATEGORY  
ORDER BY TOTAL_REVENUE DESC  
LIMIT 10;
```

SUBCATEGORY	TOTAL_REVENUE
Desktops	9906356.50
Televisions	4308719.19
Projectors & Screens	3767522.00
Water Heaters	3547822.50
Camcorders	3357990.00
Laptops	3164777.20
Movie DVD	3131006.44
Touch Screen Phones	3083462.00
Smart phones & PDAs	2805657.00
Refrigerators	2152664.41

Top 10 Products by Profit

```
-- Top 10 Product by Profit --  
SELECT PRODUCT_NAME,  
       SUM(QUANTITY * (UNIT_PRICE_USD - UNIT_COST_USD)) AS TOTAL_PROFIT  
FROM PRODUCTS  
JOIN SALES ON PRODUCTS.PRODUCTKEY = SALES.PRODUCTKEY  
GROUP BY PRODUCT_NAME  
ORDER BY TOTAL_PROFIT DESC  
LIMIT 10;
```

PRODUCT_NAME	TOTAL_PROFIT
WWI Desktop PC2.33 X2330 Black	337986.00
Adventure Works Desktop PC2.33 XD233 Silver	311663.95
Adventure Works Desktop PC2.33 XD233 Brown	310368.05
Adventure Works Desktop PC2.33 XD233 Black	299352.90
Adventure Works Desktop PC2.33 XD233 White	292225.45
WWI Desktop PC2.33 X2330 White	283908.24
WWI Desktop PC2.33 X2330 Brown	282679.20
Adventure Works 52" LCD HDTV X590 White	263727.12
Adventure Works 52" LCD HDTV X590 Black	250152.93
WWI Desktop PC2.33 X2330 Silver	240891.84

Seasonality Check

Total Orders & Revenue

2018

MONTH	ORDER_COUNT	REVENUE
January	1035	917075.35
February	1268	1344342.70
March	463	434951.15
April	95	70627.54
May	1181	1073265.56
June	1106	973975.33
July	1052	964149.46
August	1223	1026845.73
September	1334	1181765.14
October	1434	1241730.03
November	1477	1478026.63
December	2520	2082206.04

2019

MONTH	ORDER_COUNT	REVENUE
January	2154	1940899.57
February	2323	2102862.57
March	921	845925.09
April	174	149892.71
May	1918	1594446.47
June	1747	1404861.40
July	1713	1408714.62
August	1893	1500784.77
September	1909	1547870.73
October	2002	1575168.82
November	1890	1715659.88
December	2967	2477295.85

2020

MONTH	ORDER_COUNT	REVENUE
January	2457	2068951.12
February	2652	2227379.57
March	775	674990.66
April	211	217642.10
May	1107	878026.63
June	863	763883.23
July	609	512805.18
August	488	416647.08
September	409	380431.52
October	338	245647.59
November	356	256701.02
December	761	651526.44

2021

MONTH	ORDER_COUNT	REVENUE
January	617	513021.58
February	595	526266.90

From 2020, sales drops significantly lower than ever

December, January & February generates highest Sales

Lowest performance in March & April

```
SELECT
```

```
    YEAR(ORDER_DATE) AS YEAR,  
    MONTH(ORDER_DATE) AS MONTHNUMBER,  
    MONTHNAME(ORDER_DATE) AS MONTH,  
    COUNT(ORDER_NUMBER) AS ORDER_COUNT,  
    SUM(QUANTITY*UNIT_PRICE_USD) AS REVENUE
```

```
FROM SALES
```

```
JOIN PRODUCTS ON SALES.PRODUCTKEY = PRODUCTS.PRODUCTKEY
```

```
GROUP BY YEAR, MONTHNUMBER, MONTH
```

```
ORDER BY YEAR, MONTHNUMBER, MONTH;
```


RFM Segmentations

RFM Scores

```
-- RFM SCORES
WITH RFM_SCORES AS (
    SELECT
        CUSTOMERKEY,
        RECENCY,
        FREQUENCY,
        MONETARY,
        NTILE(5) OVER (ORDER BY RECENCY ASC) AS R_SCORE,
        NTILE(5) OVER (ORDER BY FREQUENCY DESC) AS F_SCORE,
        NTILE(5) OVER (ORDER BY MONETARY DESC) AS M_SCORE
    FROM (
        SELECT CUSTOMERKEY,
            DATEDIFF(CURDATE(), MAX(ORDER_DATE)) AS RECENCY,
            COUNT(DISTINCT ORDER_NUMBER) AS FREQUENCY,
            SUM(QUANTITY * UNIT_PRICE_USD) AS MONETARY
        FROM SALES
        JOIN PRODUCTS ON SALES.PRODUCTKEY = PRODUCTS.PRODUCTKEY
        GROUP BY CUSTOMERKEY
    ) AS RFM_VALUE
)
SELECT *
FROM RFM_SCORES;
```

Customers by Segments

SEGMENT	CUSTOMER_COUNT
OTHER	5160
Champions	2111
Potential Loyalists	1306
Loyal Customers	1305
Churned	1102
At Risk	903

RFM Values

```
-- RFM VALUES
WITH RFM_VALUE AS (
    SELECT CUSTOMERKEY,
        DATEDIFF(CURDATE(), MAX(ORDER_DATE)) AS RECENCY,
        COUNT(DISTINCT ORDER_NUMBER) AS FREQUENCY,
        SUM(QUANTITY * UNIT_PRICE_USD) AS MONETARY
    FROM SALES
    JOIN PRODUCTS ON SALES.PRODUCTKEY = PRODUCTS.PRODUCTKEY
    GROUP BY CUSTOMERKEY
)
SELECT *
FROM RFM_VALUE;
```

RFM Segmentations

RFM Combined Scores

```
-- RFM COMBINED SCORES --
WITH RFM_FINAL AS (
    SELECT
        CUSTOMERKEY,
        RECENCY,
        FREQUENCY,
        MONETARY,
        NTILE(5) OVER (ORDER BY RECENCY ASC) AS R_SCORE,
        NTILE(5) OVER (ORDER BY FREQUENCY DESC) AS F_SCORE,
        NTILE(5) OVER (ORDER BY MONETARY DESC) AS M_SCORE
    FROM (
        SELECT
            CUSTOMERKEY,
            DATEDIFF(CURDATE(), MAX(ORDER_DATE)) AS RECENCY,
            COUNT(DISTINCT ORDER_NUMBER) AS FREQUENCY,
            SUM(QUANTITY * UNIT_PRICE_USD) AS MONETARY
        FROM SALES
        JOIN PRODUCTS ON SALES.PRODUCTKEY = PRODUCTS.PRODUCTKEY
        GROUP BY CUSTOMERKEY
    ) AS RFM_VALUES
)
SELECT
    CUSTOMERKEY,
    CONCAT(R_SCORE, F_SCORE, M_SCORE) AS RFM_SCORE,
    R_SCORE,
    F_SCORE,
    M_SCORE
FROM RFM_FINAL
ORDER BY RFM_SCORE DESC;
```

RFM Segmentations

```
-- RFM SEGMENTATIONS
WITH RFM_SEGMENTS AS(
    SELECT
        CUSTOMERKEY,
        CONCAT(R_SCORE, F_SCORE, M_SCORE) AS RFM_SCORE,
        CASE
            WHEN R_SCORE <= 2 AND F_SCORE >= 4 AND M_SCORE >= 4 THEN 'Champions'
            WHEN R_SCORE >= 3 AND F_SCORE >= 4 AND M_SCORE >= 3 THEN 'Loyal Customers'
            WHEN R_SCORE >= 3 AND F_SCORE >= 3 AND M_SCORE >= 3 THEN 'Potential Loyalists'
            WHEN R_SCORE >= 4 AND F_SCORE BETWEEN 2 AND 3 THEN 'At Risk'
            WHEN R_SCORE = 5 AND F_SCORE <= 2 AND M_SCORE <= 2 THEN 'Churned'
            ELSE 'OTHER'
        END AS SEGMENT
    FROM (
        SELECT CUSTOMERKEY,
            NTILE(5) OVER (ORDER BY RECENCY ASC) AS R_SCORE,
            NTILE(5) OVER (ORDER BY FREQUENCY) AS F_SCORE,
            NTILE(5) OVER (ORDER BY MONETARY) AS M_SCORE
        FROM (
            SELECT CUSTOMERKEY,
                DATEDIFF(CURDATE(), MAX(ORDER_DATE)) AS RECENCY,
                COUNT(DISTINCT ORDER_NUMBER) AS FREQUENCY,
                SUM(QUANTITY * UNIT_PRICE_USD) AS MONETARY
            FROM SALES
            JOIN PRODUCTS ON SALES.PRODUCTKEY = PRODUCTS.PRODUCTKEY
            GROUP BY CUSTOMERKEY
        )
    ) AS RFM_VALUE
) AS RFM_SCORE
)
SELECT *
FROM RFM_SEGMENTS
ORDER BY SEGMENT;
```

Customers by Segments

```
-- COUNT OF CUSTOMERS BY SEGMENTS
WITH RFM_SEGMENTS AS(
    SELECT
        CUSTOMERKEY,
        CONCAT(R_SCORE, F_SCORE, M_SCORE) AS RFM_SCORE,
        CASE
            WHEN R_SCORE <= 2 AND F_SCORE >= 4 AND M_SCORE >= 4 THEN 'Champions'
            WHEN R_SCORE >= 3 AND F_SCORE >= 4 AND M_SCORE >= 3 THEN 'Loyal Customers'
            WHEN R_SCORE >= 3 AND F_SCORE >= 3 AND M_SCORE >= 3 THEN 'Potential Loyalists'
            WHEN R_SCORE >= 4 AND F_SCORE BETWEEN 2 AND 3 THEN 'At Risk'
            WHEN R_SCORE = 5 AND F_SCORE <= 2 AND M_SCORE <= 2 THEN 'Churned'
            ELSE 'OTHER'
        END AS SEGMENT
    FROM (
        SELECT CUSTOMERKEY,
            NTILE(5) OVER (ORDER BY RECENCY ASC) AS R_SCORE,
            NTILE(5) OVER (ORDER BY FREQUENCY) AS F_SCORE,
            NTILE(5) OVER (ORDER BY MONETARY) AS M_SCORE
        FROM (
            SELECT CUSTOMERKEY,
                DATEDIFF(CURDATE(), MAX(ORDER_DATE)) AS RECENCY,
                COUNT(DISTINCT ORDER_NUMBER) AS FREQUENCY,
                SUM(QUANTITY * UNIT_PRICE_USD) AS MONETARY
            FROM SALES
            JOIN PRODUCTS ON SALES.PRODUCTKEY = PRODUCTS.PRODUCTKEY
            GROUP BY CUSTOMERKEY
        )
    ) AS RFM_VALUE
) AS RFM_SCORE
)
SELECT
    SEGMENT,
    COUNT(CUSTOMERKEY) AS CUSTOMER_COUNT
FROM RFM_SEGMENTS
GROUP BY SEGMENT
ORDER BY CUSTOMER_COUNT DESC;
```

What types of products does the company sell, and where are customers located?

GE sell

Products

CATEGORY
Audio
TV and Video
Computers
Cameras and camcorders
Cell phones
Music, Movies and Audio Books
Games and Toys
Home Appliances

To

Country

Country	Customer_Count
United States	6828
United Kingdom	1944
Canada	1553
Germany	1473
Australia	1420
Netherlands	733
France	670
Italy	645

With

Total Orders

Country	TOTAL_ORDERS
United States	14221
United Kingdom	3421
Germany	2440
Canada	2281
Australia	1182
Italy	1101
Netherlands	976
France	704

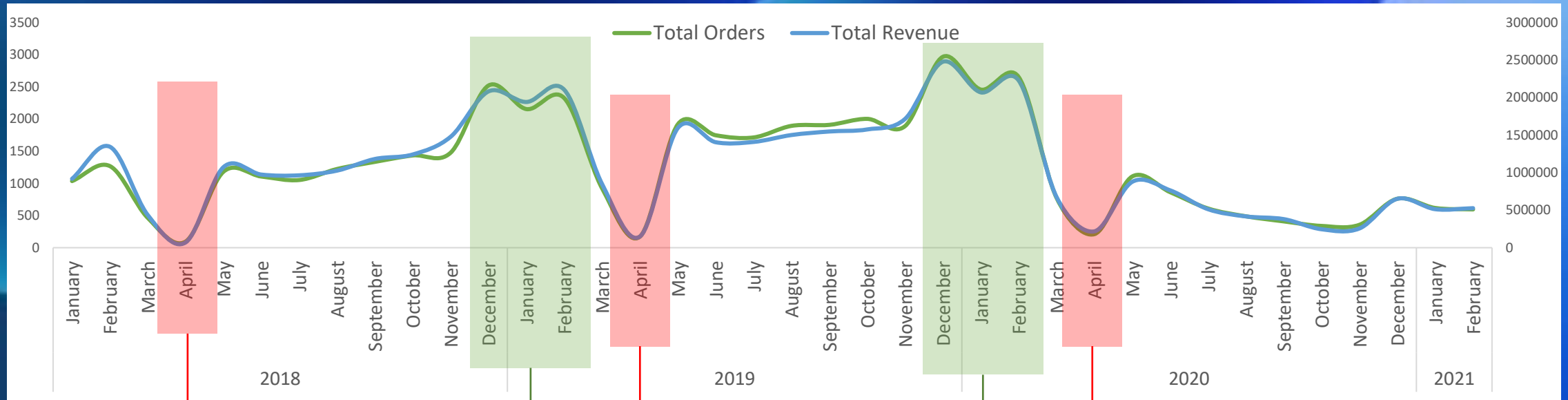
Top 10
Subcategories
by Revenue

SUBCATEGORY	TOTAL_REVENUE
Desktops	9906356.50
Televisions	4308719.19
Projectors & Screens	3767522.00
Water Heaters	3547822.50
Camcorders	3357990.00
Laptops	3164777.20
Movie DVD	3131006.44
Touch Screen Phones	3083462.00
Smart phones & PDAs	2805657.00
Refrigerators	2152664.41

Top 10
Products by
Revenue

PRODUCT_NAME	TOTAL_PROFIT
WWI Desktop PC2.33 X2330 Black	337986.00
Adventure Works Desktop PC2.33 XD233 Silver	311663.95
Adventure Works Desktop PC2.33 XD233 Brown	310368.05
Adventure Works Desktop PC2.33 XD233 Black	299352.90
Adventure Works Desktop PC2.33 XD233 White	292225.45
WWI Desktop PC2.33 X2330 White	283908.24
WWI Desktop PC2.33 X2330 Brown	282679.20
Adventure Works 52" LCD HDTV X590 White	263727.12
Adventure Works 52" LCD HDTV X590 Black	250152.93
WWI Desktop PC2.33 X2330 Silver	240891.84

Are there any seasonal patterns or trends for order volume or revenue?



→ Every year, **Orders & Revenue** start to decline from **March**, **lowest** in **April** and returning to **normal** from **May**. This means that there is a **seasonal decline**.

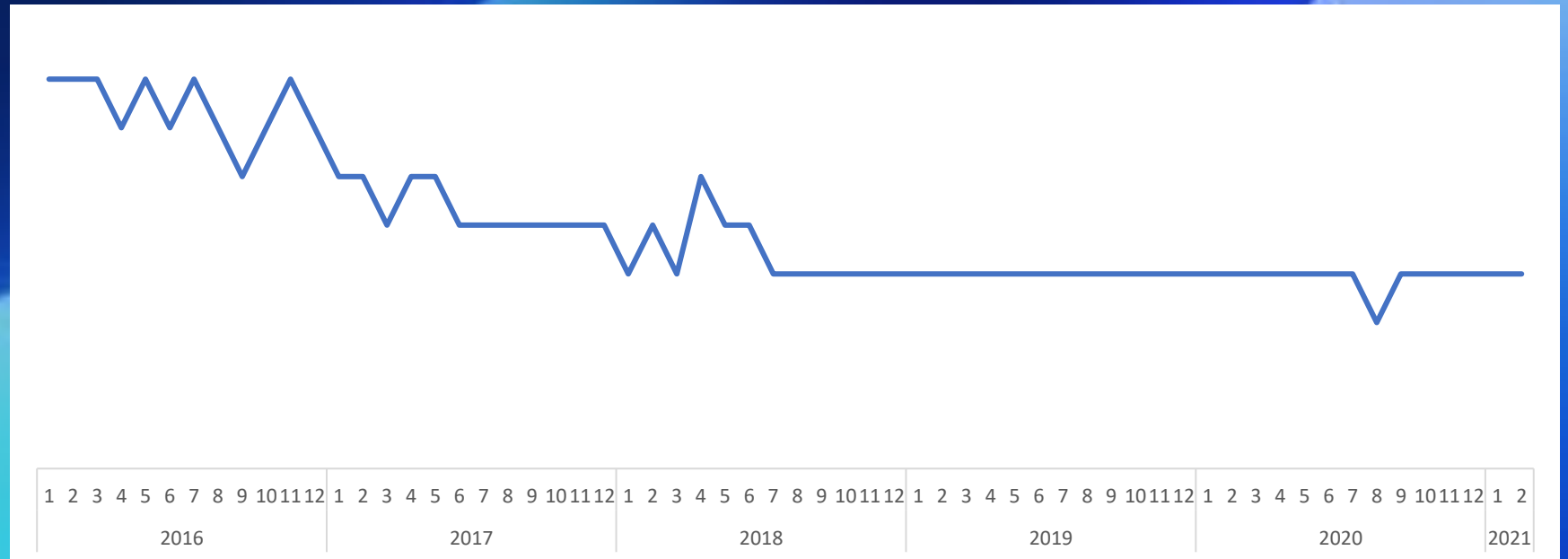
→ On the other hand, **Orders & Revenue** **increases** from the **end of October**, which is **highest** in **December, January, and February** compared to the entire year.

How long is the average delivery time in days? Has that changed over time?

Average delivery time

AVG_DELIVERY_TIME
5

Average delivery time for the past 3 years, has been maintained at 5 days.



Is there a difference in average order value (AOV) for online vs. in-store sales?

Average Order Value

SALES_CHANNELS	AOV
IN-STORE	2137.82
ONLINE	2043.79

Total Orders by Channels

Sales_Channel	Total_Orders	Percentage
IN-STORE	20746	78.80
ONLINE	5580	21.20

→ AOV is pretty much similar but the difference between Sales Channels are Noticeable

The background features several overlapping, curved, organic shapes in various shades of blue and teal. The colors range from a deep, dark blue to a bright, light cyan. The shapes are layered, creating a sense of depth and movement. The overall composition is abstract and modern.

Thank You.